Rithesh Kumar

Member of Technical Staff, OpenAI
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RESEARCH FOCUS

I work on building realtime interface to AGI at OpenAI. My work spans voice assistants, generative audio, speech synthesis, diffusion models, efficient distillation and text-based audio editing. My research has been rooted in the fundamentals of generative modeling and deep learning since 2016.

EXPERIENCE

Member of Technical Staff, OpenAI 2025-present
Building next-generation voice interfaces to AGI.

Senior Research Scientist, Adobe Research 2023-2025
Led speech generation research for controllable text-to-speech synthesis, automatic dubbing, and speech editing. Shipped core models behind Adobe Firefly Translate Video and Audio, Text-to-Avatar, and Adobe TTS API. Developed broadcast-quality audio diffusion models and efficient distillation algorithms for voice translation and controllable TTS across 25+ languages and dialects.

Technical Lead, Audio Research, Descript (prev. Lyrebird) 2019-2023
Led audio research across neural text-to-speech, voice cloning, speech editing, and audio language modeling. Shipped 4+ research models powering Overdub, Regenerate, and AI Voices. Developed audio language models and high-fidelity voice cloning systems for text-based speech correction and 44.1 kHz voice generation.

SHIPPED PRODUCTS

Adobe Firefly Translate Video and Audio, Text-to-Avatar 2025
Speaker-preserving voice translation for multilingual media workflows, plus voice generation for animated avatars.

Descript Regenerate 2023
Speech editing and audio repair that turns awkward cuts, gaps, and retakes into smooth generated speech.

Descript AI Voices / Overdub 2018
Voice cloning and text-to-speech systems behind Overdub and later AI voice creation workflows.

EDUCATION

M.Sc., Computer Science, Mila - Universite de Montreal 2017-2019
CGPA: 4.15 / 4.3.

Supervised by Prof. Yoshua Bengio. Thesis: Maximum Entropy Generators for Energy-Based Models.

B.E., Computer Science and Engineering, Anna University 2013-2017
CGPA: 8.63 / 10.0

Rank: 46 among 16,449 candidates.

PUBLICATIONS

ICML

[1] William Chen, Prem Seetharaman, Rithesh Kumar, Oriol Nieto, Shinji Watanabe, Justin Salamon, Zeyu Jin. "AudioChat: Unified Audio Storytelling, Editing, and Understanding with Transfusion Forcing." *ICML 2026*.

[2] Yinghao Aaron Li, Rithesh Kumar, Zeyu Jin. "DMOSpeech: Direct Metric Optimization via Distilled Diffusion Model in Zero-Shot Speech Synthesis." *ICML 2025*.

ICLR

[3] Justin Lovelace, Rithesh Kumar, Jiaqi Su, Ke Chen, Kilian Q. Weinberger, Zeyu Jin. "SpeechOp: Inference-Time Task Composition for Generative Speech Processing." *ICLR 2026*.

[4] Max Morrison, Rithesh Kumar, Kundan Kumar, Prem Seetharaman, Aaron Courville, Yoshua Bengio. "Chunked Autoregressive GAN for Conditional Waveform Synthesis." *ICLR 2022*.

[5] Soroush Mehri, Kundan Kumar, Ishaan Gulrajani, Rithesh Kumar, Shubham Jain, Jose Sotelo, Aaron Courville, Yoshua Bengio. "SampleRNN: An Unconditional End-to-End Neural Audio Generation Model." *ICLR 2017*.

NeurIPS

[6] Rithesh Kumar, Prem Seetharaman, Alejandro Luebs, Ishaan Kumar, Kundan Kumar. "High-Fidelity Audio Compression with Improved RVQGAN." *NeurIPS 2023*. Spotlight.

[7] Kundan Kumar, Rithesh Kumar, Thibault de Boissiere, Lucas Gestin, Wei Zhen Teoh, Jose Sotelo, Alexandre de Brebisson, Yoshua Bengio, Aaron Courville. "MelGAN: Generative Adversarial Networks for Conditional Waveform Synthesis." *NeurIPS 2019*.

ICASSP

[8] Prem Seetharaman*, Rithesh Kumar*. "Taming Audio VAEs via Target-KL Regularization." *ICASSP 2026*. Equal contribution.

[9] Yutong Wen, Ke Chen, Prem Seetharaman, Oriol Nieto, Jiaqi Su, Rithesh Kumar, Minje Kim, Paris Smaragdis, Zeyu Jin, Justin Salamon. "PromptSep: Generative Audio Separation via Multimodal Prompting." *ICASSP 2026*.

[10] Heitor R. Guimaraes, Jiaqi Su, Rithesh Kumar, Tiago H. Falk, Zeyu Jin. "DiTSE: High-Fidelity Generative Speech Enhancement via Latent Diffusion Transformers." *ICASSP 2026*.

ISMIR

[11] Hugo Flores Garcia, Prem Seetharaman, Rithesh Kumar, Bryan Pardo. "VampNet: Music Generation via Masked Acoustic Token Modeling." *ISMIR 2023*.

WASPAA

[12] Yunyun Wang, Jiaqi Su, Adam Finkelstein, Rithesh Kumar, Ke Chen, Zeyu Jin. "DiTVC: One-Shot Voice Conversion via Diffusion Transformer with Environment and Speaking Rate Cloning." *WASPAA 2025*.

CHI

[13] Stephen Brade, Sam Anderson, Rithesh Kumar, Zeyu Jin, Anh Truong. "SpeakEasy: Enhancing Text-to-Speech Interactions for Expressive Content Creation." *CHI 2025*.

Thesis / Preprints / Other

[14] Sonal Kumar, Prem Seetharaman, Ke Chen, Oriol Nieto, Jiaqi Su, Zhepei Wang, Rithesh Kumar, Dinesh Manocha, Nicholas J. Bryan, Zeyu Jin, Justin Salamon. "TAC: Timestamped Audio Captioning." *2026*.

[15] Rithesh Kumar, Kundan Kumar, Vicki Anand, Yoshua Bengio, Aaron Courville. "NU-GAN: High resolution neural upsampling with GAN." *2020*.

[16] Rithesh Kumar, Sherjil Ozair, Anirudh Goyal, Aaron Courville, Yoshua Bengio. "Maximum Entropy Generators for Energy-Based Models." *M.Sc. Thesis, 2019*.

[17] Kyle Kastner, Rithesh Kumar, Tim Cooijmans, Aaron Courville. "Harmonic Recomposition using Conditional Autoregressive Modeling." *2018*.

[18] Rithesh Kumar, Jose Sotelo, Kundan Kumar, Alexandre de Brebisson, Yoshua Bengio. "ObamaNet: Photo-realistic lip-sync from text." *2017*.